



Indian Institute of Technology Ropar
Department of Mathematics
MA302
Optimization methods with Applications
Semester - 2 of Academic Year 2025-2026

Problem Sheet - 2

1. Write the following LPPs into the standard form.

(a)

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + x_2 \\ &\text{subject to } \quad 2x_1 + x_2 \leq 4, \\ &\quad \quad \quad -x_1 + x_2 \geq 8 \\ &\quad \quad \quad x_1, x_2 \geq 0. \end{aligned}$$

(b)

$$\begin{aligned} &\text{Minimize } Z = 2x_1 + x_2 + 4x_3 \\ &\text{subject to } \quad -2x_1 + 4x_2 \leq 4, \\ &\quad \quad \quad x_1 + 2x_2 + x_3 \geq 5 \\ &\quad \quad \quad 2x_1 + 3x_3 \leq 2 \\ &\quad \quad \quad x_1, x_2 \geq 0 \\ &\text{and } x_3 \text{ is unrestricted in sign.} \end{aligned}$$

2. Without sketching the graph find all vertices of the set of feasible solutions of

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + 4x_2 \\ &\text{subject to } \quad 3x_1 + 5x_2 \leq 15, \\ &\quad \quad \quad 3x_1 + 2x_2 \leq 12, \\ &\quad \quad \quad x_1, x_2 \geq 0. \end{aligned}$$

3. Let the function $f(x_1, x_2) = -3x_1 + x_2$ be minimized over the solid triangle ABC with vertices $A(-1, 0)$, $B(2, 0)$, $C(0, 1)$. Formulate this as LPP and solve the same graphically.
4. Identify all basic feasible solutions for the system. Also classify them as Degenerate or Non - denegerate basic feasible solutions.

$$\begin{aligned} &x_1 + 4x_2 + x_3 = 8, \\ &x_1 + 2x_2 + x_4 = 4, \\ &x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

5. Solve the following LPPs using algebraic method.

(a)

$$\begin{array}{ll}\text{Maximize } Z = 2x_1 + 4x_2 \\ \text{subject to } & 3x_1 + 5x_2 \leq 15, \\ & 3x_1 + 2x_2 \leq 12, \\ & x_1, x_2 \geq 0.\end{array}$$

(b)

$$\begin{array}{ll}\text{Maximize } Z = 2x_1 + 5x_2 \\ \text{subject to } & x_1 + 2x_2 \leq 20, \\ & x_1 + x_2 \leq 15, \\ & x_1, x_2 \geq 0.\end{array}$$

(c)

$$\begin{array}{ll}\text{Maximize } Z = x_1 + 2x_2 \\ \text{subject to } & x_1 + x_2 \leq 8, \\ & 2x_1 + x_2 \leq 8, \\ & x_1, x_2 \geq 0.\end{array}$$

★ ★ ★ ★ **End** ★ ★ ★ ★